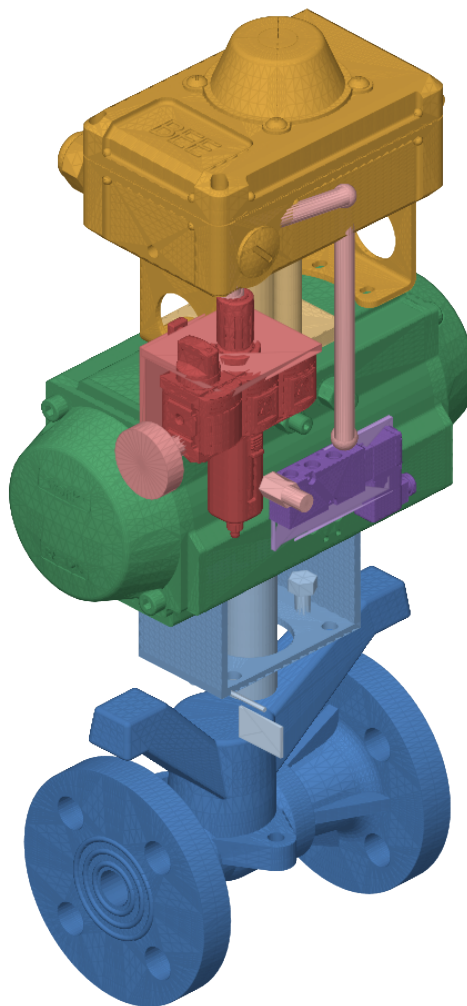


ASSEMBLY MANUAL

Configuration 1 · 15 mm L&T ball valve automation package

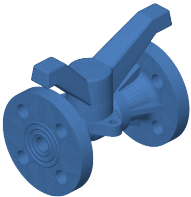
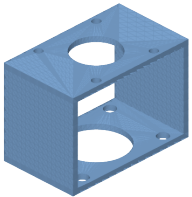
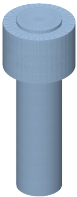
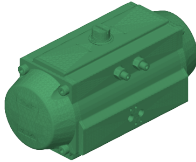
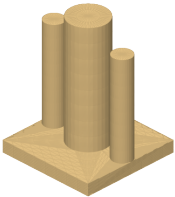
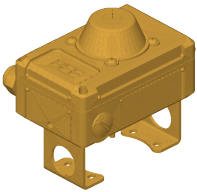
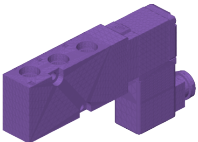
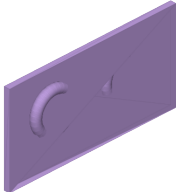
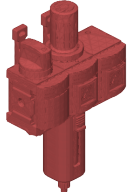
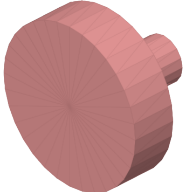
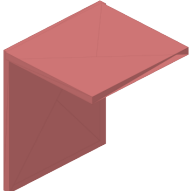
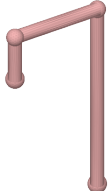
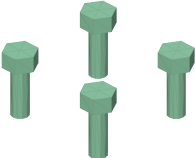
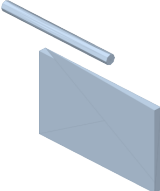
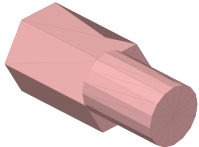
BV1121 - PA1019 - TP0026 - PN00

FAIL-TO-OPEN · SPRING RETURN



Parts

One colour per component group; lighter shades are that group's minor parts.
Parts marked † are schematic stand-ins pending supplier CAD.

 Ball valve 15 mm BV1121 · L&T L1RF1C 1x	 Mounting bracket BK1001 · Rajdeep 1x	 Drive coupling BK1001 · Rajdeep 1x	 Pneumatic actuator PA1019 · spring return 1x	 LSB riser bracket VDI/VDE 3845 1x
 Limit switch box LB · Valmet KS2V 1x	 Solenoid valve SV · Rotex 30318 NAMUR 1x	 NAMUR plate + O-rings † rubber adapter 1x + 2x	 Air filter regulator FR1001 · Shavo SB10 1x	 Dial gauge † AFR gauge port 1x
 AFR L-plate † slotted 1x	 Air pipe † bent 1/4" + connectors 1x	 Bolt/nut sets † placeholder sizes see steps	 Nameplate tag salvaged in step 2 1x	 Supply fitting † push-in, 1/4" 1x

FASTENERS (placeholder sizes, V1)

2x M6 socket-head (bracket) · 4x M5 studs + nuts (LSB) · 2x M5 (AFR plate) · 2x M6 hex (L-plate to actuator)
2x M5 long Allen + blanking nuts + silencer (SOV) · 1/4" connectors + PTFE tape (air pipe) · 4x M6 + 2x M6 + washers (final bolt-up)

How this manual works



- COMBINE 1

AFR unit → actuator (step 11)
- COMBINE 2

SOV → actuator (step 12)
- COMBINE 3

air pipe AFR → SOV (step 13)
- COMBINE 4

dressed actuator → valve (step 14)
- FINISH

bolt-up · tag · supply fitting (steps 15-16)

COLOUR KEY & READING THE PICTURES

- Valve group

valve · bracket · coupling · tag · valve bolts · lighter shade = minor part of the group
- Actuator

actuator body · base bolts · lighter shade = minor part of the group
- Limit switch box

LSB · riser bracket · lighter shade = minor part of the group
- Air set (AFR)

AFR · gauge · L-plate · air pipe · supply fitting · lighter shade = minor part of the group
- Solenoid valve

SOV · NAMUR plate + O-rings · lighter shade = minor part of the group

→ arrow = direction the part moves (or comes off, for removal steps) ■ every check box must pass before the next step

Testing & QC (stroke set, cam set, function test, hydro 30 bar, seat leakage 6 bar, final QC) are round 2 — not part of this manual.

ASSEMBLY STEPS

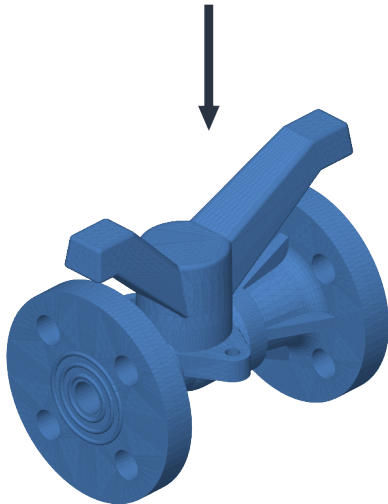
1

A1

SUB-ASSEMBLY A · VALVE

Check & clamp the valve

Check open/close. Clamp the valve in the vice, X-Y, flanges to the line.

PART(S)
Valve BV1121FASTENER
—TOOL
Bench viceTORQUE
—

CHECK

- ☐ Valve cycles freely
- ☐ Clamped square
- ☐ Bore horizontal

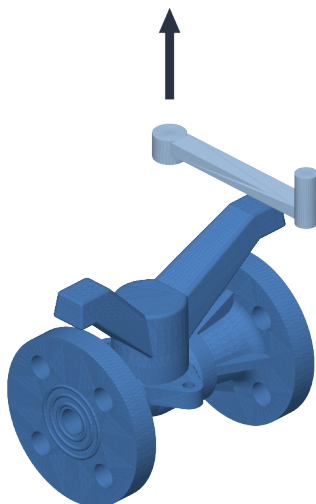
2

A2

SUB-ASSEMBLY A · VALVE

Remove lever handle + nameplate

Take off the lever and nameplate. KEEP the nameplate. Refit the stem nut.

PART(S)
Lever (off)FASTENER
1x lever nut/bolt; stem nutTOOL
Ring spannerTORQUE
—

CHECK

- ☐ Nameplate retained for tagging
- ☐ Note ball position (closed)

ASSEMBLY STEPS

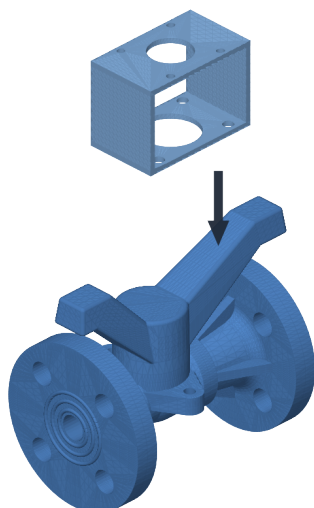
3

A3

SUB-ASSEMBLY A · VALVE

Fit the mounting bracket

Fit the bracket over the stem onto the stop-plate ears.



PART(S)

Bracket BK1001

FASTENER

2x M6 socket-head

TOOL

Allen key 5 mm

TORQUE

~10 Nm

CHECK

☐ Bracket flat, no rock☐ ISO 5211 pattern up

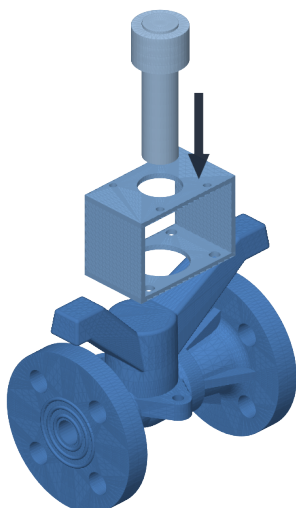
4

A4

SUB-ASSEMBLY A · VALVE

Fit the drive coupling

Drop the drive coupling/adaptor onto the stem.



PART(S)

Coupling BK1001

FASTENER

None (press fit)

TOOL

Hand

TORQUE

—

CHECK

☐ Coupling seated on stem flats☐ Square tang up

ASSEMBLY STEPS

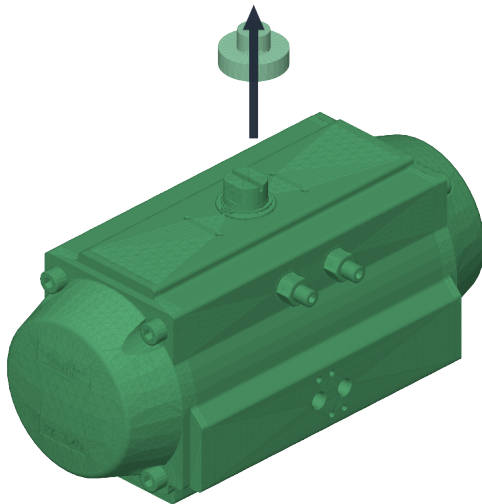
5

B1

SUB-ASSEMBLY B · ACTUATOR

Remove the actuator top cap

Remove the top cap / position indicator. Wipe clean.



PART(S)

Indicator (off)

FASTENER

—

TOOL

Screwdriver

TORQUE

—

CHECK

☐ NAMUR pinion drive exposed

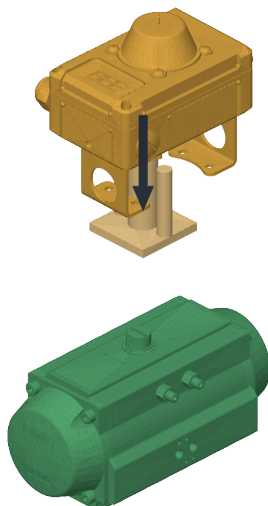
6

B2

SUB-ASSEMBLY B · ACTUATOR

Fit riser + limit switch box

Fit the LSB riser bracket and switch box on the actuator top. Engage the shaft.



PART(S)

LSB riser + LSB

FASTENER

4x M5 studs + nuts

TOOL

Hand + spanner

TORQUE

~5 Nm

CHECK

☐ LSB shaft engaged in pinion adapter BEFORE nuts pulled☐ Box square

ASSEMBLY STEPS

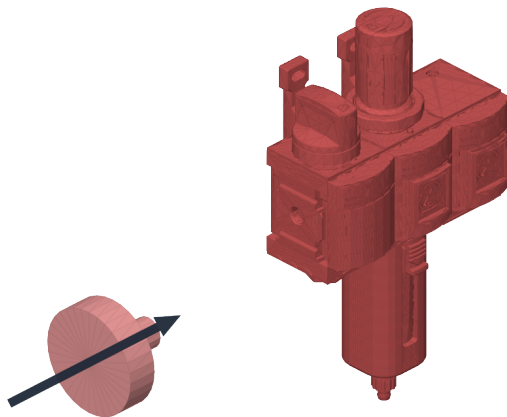
7

C1

SUB-ASSEMBLY C · AFR

Thread in the dial gauge

Thread the dial gauge into the AFR gauge port.



PART(S)

AFR + gauge

FASTENER

Gauge threaded

TOOL

Hand

TORQUE

—

CHECK

☐ PTFE on thread☐ Gauge readable from front

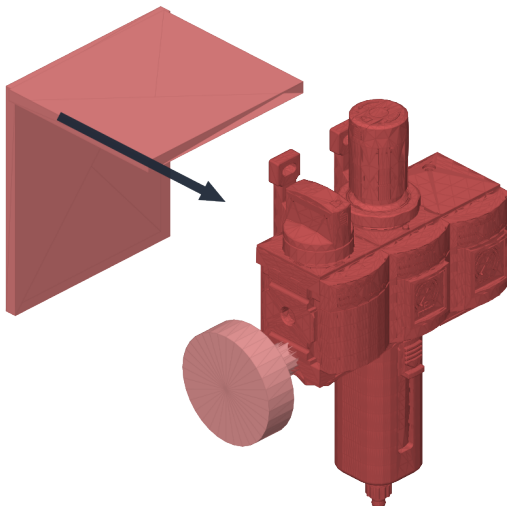
8

C2

SUB-ASSEMBLY C · AFR

Bolt AFR to its L-plate

Bolt the AFR body to the L-plate.



PART(S)

AFR L-plate

FASTENER

2x M5 nuts/bolts

TOOL

Spanner

TORQUE

~6 Nm

CHECK

☐ Bowl points down☐ Slots vertical for height adjust

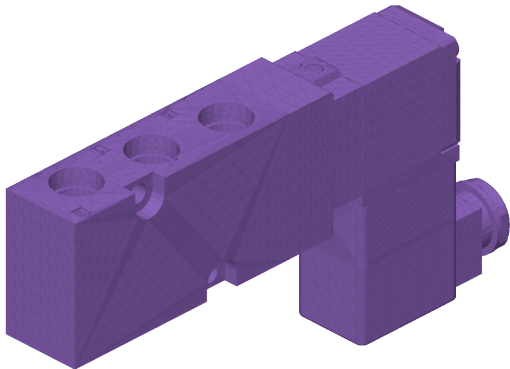
9

SUB-ASSEMBLY D · SOV

Select SOV, set the O-ring gate

Select 3/2 or 5/2 per customer. Set the O-ring gate to the required port.

D1



PART(S)
SOV SV (Rotex 30318)

FASTENER
—

TOOL
—

TORQUE
—

CHECK
☐ Gate matches ordered function

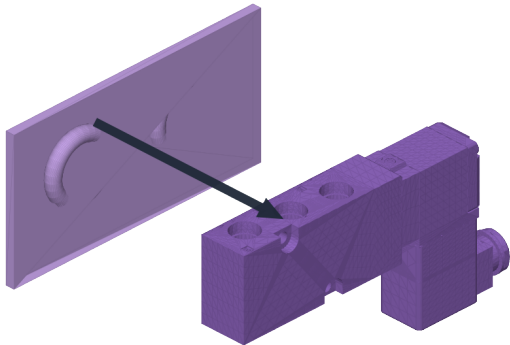
10

SUB-ASSEMBLY D · SOV

Fit NAMUR plate + O-rings

Fit the rubber NAMUR plate and 2 O-rings. Align the O-rings to the SOV.

D2



PART(S)
NAMUR plate + 2 O-rings

FASTENER
—

TOOL
Hand

TORQUE
—

CHECK
☐ O-rings seated
☐ Protective film peeled

ASSEMBLY STEPS

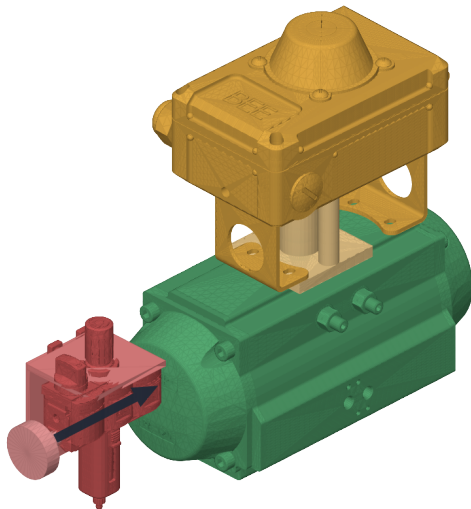
11

E1

COMBINE 1 · C → B

Bolt AFR unit to the actuator

Bolt the AFR L-plate to the actuator side face.



PART(S)

AFR unit → actuator

FASTENER

2x M6 hex into top threading

TOOL

Open-end spanner

TORQUE

~8 Nm

CHECK

- ☐ Plate square
- ☐ AFR height set so tube run will line up

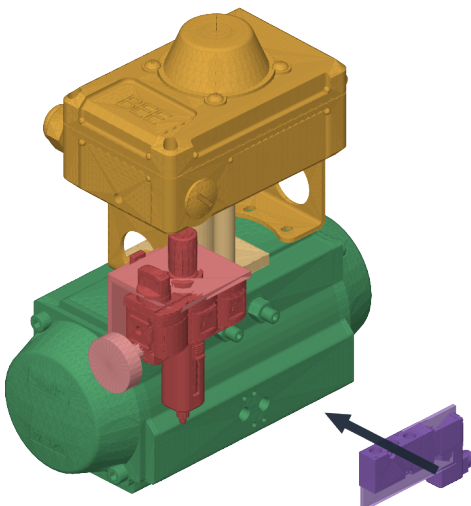
12

F1

COMBINE 2 · D → B+C

Mount SOV on the NAMUR face

Offer the SOV (with plate) to the actuator NAMUR face; bolt through the O-ring holes. Blank all air holes except the middle (supply). Fit the silencer.



PART(S)

SOV → actuator

FASTENER

2x M5 long Allen; blanking nuts

TOOL

Ball-end Allen key

TORQUE

~6 Nm

CHECK

- ☐ O-rings compressed evenly
- ☐ Supply hole open
- ☐ Exhaust points clear

ASSEMBLY STEPS

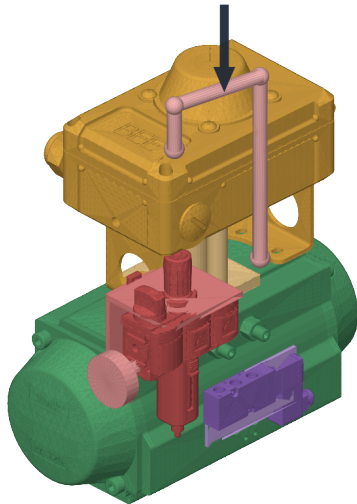
13

G1

COMBINE 3 · AIR TRAIN

Fit the air pipe (AFR → SOV)

AFR "in" open (supply), elbow on "out". Fit the bent pipe from AFR to the SOV middle hole.



PART(S)

Air pipe

FASTENER

1/4" connectors; PTFE tape

TOOL

Open-end spanner

TORQUE

—

CHECK

☐ Pipe geometry matches AFR height☐ Nuts not cross-threaded

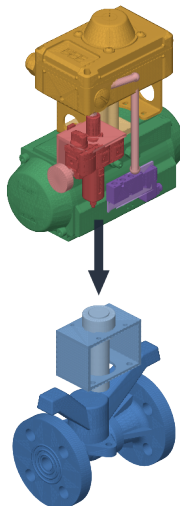
14

H1

COMBINE 4 · ONTO VALVE

Lower dressed actuator onto valve

Lower the fully dressed actuator as ONE unit onto the bracket. Engage stem-to-pinion drive.



PART(S)

Actuator unit → valve

FASTENER

—

TOOL

Hand

TORQUE

—

CHECK

☐ Coupling tang aligned to pinion — rotate coupling, never force☐ Fail-action orientation correct

ASSEMBLY STEPS

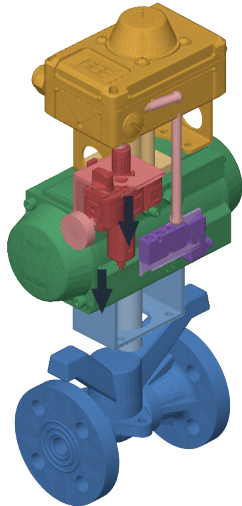
15

H2

FINISH

Final bolt-up

Bolt actuator base to bracket, then tighten bracket-to-valve.



PART(S)

Bolt/nut sets

FASTENER

4x M6 base + 2x M6 valve; washers

TOOL

Allen key + spanner

TORQUE

Base ~20 Nm · valve ~10 Nm

CHECK

- ☐ Cross-tighten base
- ☐ Actuator square before final torque

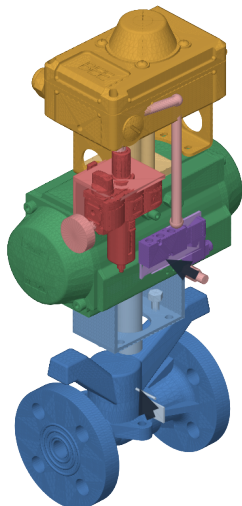
16

H3

FINISH

Tag + air supply, set 4 bar

Re-tie the salvaged nameplate tag. Fit the air-supply push-in fitting to the AFR inlet. Set the regulator to 4 bar.



PART(S)

Tag + supply fitting

FASTENER

—

TOOL

Wire; hand

TORQUE

—

CHECK

- ☐ Spec tag on unit
- ☐ Supply fitting PTFE-taped
- ☐ AFR set to 4 bar

Deferred to round 2 (not in this manual)

- Stroke / end-stop set
- LSB cam set
- Function test (24 VDC, cycle, no leaks)
- Body hydro 30 bar water (ISO 5208)
- Seat leakage 6 bar air
- Final QC

Built from ValveOS/docs/golden-steplist-config01.md (FINAL v1, decisions locked 2026-07-13).

Illustrations rendered from the Config 1 .glb parts (ValveOS/app/public). Parts marked † are schematic stand-ins.